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Review

Effect of promoters and reactant concentration on the selective oxidation of n-butane to maleic anhydride using vanadium phosphorus oxide catalysts

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Abstract

The role of promoters and reactant concentration on the selective oxidation of n-butane to maleic anhydride is reviewed for vanadium phosphorus oxide catalysts. It is considered that the promoters have a twofold structural role: (a) to enable the formation of the required VPO compounds and decrease the formation of deleterious phases, and (b) to enable the formation of solid solutions which regulate the catalytic activity of the solid. Studies on the effect of reactant concentration have indicated that considerable advantages can be gained for maleic anhydride yields by using increased n-butane concentrations, but as yet few studies have been made in this area.

Keywords: vanadium-phosphorus oxide, maleic anhydride, butane oxidation, promoters.

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1. INTRODUCTION

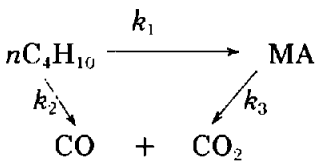
There has been considerable interest in the selective oxidation of n-butane to maleic anhydride since Bergman and Frisch [1] disclosed in 1966 that this reaction could be catalysed by vanadium phosphorus mixed oxides. At that time benzene was the preferred feedstock for maleic anhydride production and n-butane was uncompetitive due to the comparatively lower yields that could be obtained. However, the large scale exploitation of natural gas brought the ready availability in the USA and northern Europe of large economic supplies of n-butane as a natural gas liquid. This, together with the imposition of strict pollution control measures on atmospheric benzene emissions in the USA, meant that by the end of the 1970's n-butane became the preferred feedstock for most plant locations. Consequently, a large part of the older benzene based maleic anhydride plants have been either closed or converted to n-butane and new production capacity is expected to be based on n-butane.

It is against this background that a considerable number of patent and open literature reports have been made since the initial disclosure. The patent literature has concentrated on improving the catalyst performance by inclusion of promoter compounds or by using novel preparation procedures. The open literature, however, has tended to concentrate on the characterisation of the vanadium phosphorus oxide compounds and the study of well defined compounds, and very few systematic studies of promoter effects have been reported. Hodnett [2] has provided an extensive review of the literature concentrating on catalyst preparation and mechanistic aspects of the reaction, and these areas will not be duplicated in this article. In addition, a recent series of reviews [3–8] have covered a wide range of material, including reactor technology, reaction mechanism and catalyst preparations for this process. Consequently, the purpose of this review is to provide an overall view of the promotion of vanadium phosphorus oxide catalysts for selective n-butane oxidation, and the majority of the reviewed literature will necessarily be based on patents. In addition, recent literature has indicated that the n-butane concentration can significantly effect the overall maleic anhydride yield and this area will also be covered in the present review.

2. COMPARISON OF CATALYST PERFORMANCE

Since it is the aim of this review to provide an overview of promoter effects, it is necessary to compare the data presented by different researchers for different formulations. The results presented for the greater majority of catalysts for this reaction are obtained using fixed bed laboratory reactors with hydrocarbon concentrations below the lower explosive limit (< 1.5 mol-% n-butane in air). The results are typically quoted as n-butane conversions, maleic anhydride selectivities and pass yields for a range of reaction conditions. How-

ever, direct comparison of results between different publications can be difficult because different ranges of reaction conditions are usually encountered. To overcome this and obtain a quantitative comparison of the various catalysts, it is necessary to obtain some measure of each catalyst's specific activity. When using inlet n-butane concentrations below 2 mol-%, carbon monoxide and carbon dioxide are the only significant carbon containing by-products. The reaction data can then be satisfactorily accounted by a simple series-consecutive pseudo first order reaction scheme [9]:



Using this model of first order reactions in a plug flow tubular reactor an expression for the rate constant for n-butane loss (k_1+k_2) can be derived: $k_1+k_2=\text{GHSV}\cdot\ln[1/(1-x)]$, where GHSV = gas hourly space velocity and x = fractional conversion. When the rate constant for butane loss is corrected for surface area of the catalyst [i.e. $(k_1+k_2)/\text{sa}$] this factor gives a measure of the specific catalytic activity which can be used, together with selectivity data, to compare the performance of different catalysts. Unfortunately, the data concerning catalyst surface area is not available in many cases and then comparison has to be made on the basis of conversion, selectivities and pass yields alone.

3. CATALYST PREPARATION AND STRUCTURE

Vanadium phosphorus and oxygen can form a large number of distinct compounds which have been well characterised, e.g. α-$VOPO_4$ [10], $VPO_5\cdot 2H_2O$ [11], β-$VOPO_4$ [12], γ-$VOPO_4$ [13,14], $VOHPO_4$ [13–16], and $(VO)_2P_2O_7$ [13], which exhibits a similar X-ray diffraction pattern to that denoted the B phase in a number of patents, $VO(PO_3)_2$ [17], and $VO(H_2PO_4)_2$ [18]. A number of these have been studied as catalysts for selective n-butane oxidation and the inter-relationship between these compounds has been well covered in the reviews of Hodnett [2] and Bordes [4], and also in studies by Bordes and Courtine [19] and Garbassi et al. [20]. It is interesting to note that vanadium phosphorus catalysts have recently been shown to be active and selective for propene and propane partial oxidation [21–23], and therefore the application of these catalysts to a range of partial oxidation reactions can be expected in future developments. Vanadium phosphorus oxide catalysts have V : P atomic ratios typically in the range 1 : 1–1 : 1.2 and average vanadium oxidation states in the range 4.1–4.3. The influence of these parameters on catalytic perform-

ance have also been dealt with by Hodnett [2] and discussion of these points will not be duplicated here. However, to understand promoter effects, it is necessary to discuss the relationship between the method of preparation and catalyst structure so that the promoter effects on these specific compounds/phases can be considered, and therefore some comments on this area will be made to compliment material covered by Hodnett [2]. Although recent work by a number of groups has demonstrated that the VPO system can produce a wide range of different structures well in excess of those detailed above, only a relatively few VPO phases have been examined with respect to the addition of promoter compounds. Hence, this discussion solely deals with the main preparative methods that have been used to produce promoted VPO catalysts.

The main method of preparation of active catalysts is the hydrochloric acid digestion of V_2O_5 and H_3PO_4 in either an aqueous solvent [24] or non aqueous solvent, e.g. methanol, tetrahydrofuran (THF) or isobutanol [25,26], followed by solvent removal to give what is termed the catalyst precursor. This is then activated by heating, typically in the n-butane/air reaction mixture, to give the final catalyst. The role of the hydrochloric acid is apparently to reduce V_2O_5 to $VOCl_2$. A range of other reductants have been studied including lactic acid [27], oxalic acid [28], benzyl alcohol [29,30] and allyl alcohol [31]. In addition, alternative sources of the phosphorus have also been cited, e.g. $H_4P_2O_7$ [32], H_3PO_3 [33,34] and $POCl_3$ [35]. Usually the hydrochloric acid method gives superior catalytic performance [1,28] and is the preferred method for most promoted catalysts described in the patent literature. More recently [29,30] it has been shown that benzyl alcohol and hydrochloric acid as reductants yield catalysts of similar efficacy, but as yet few studies on promotion of the benzyl alcohol prepared materials have been reported. The influence of the solvent in the hydrochloric acid digestion method was initially studied by Schneider [25,26], but identification of the compounds prepared by this method has only recently proved successful [13,15]. The main precursor compound prepared for both aqueous and non aqueous solvents has been shown to be $VOHPO_4 \cdot 0.5H_2O$ which is converted to the active catalytic form of $(VO)_2P_2O_7$ on heating. However, Schneider initially showed that the role of the solvent was crucial in obtaining the best catalytic performance, and non-aqueous solvents were preferred, since when used the catalysts exhibited high surface areas ($>20\ m^2\ g^{-1}$) compared to aqueous preparations ($2–3\ m^2\ g^{-1}$). Recent studies [36,37] have confirmed the original observation of Schneider concerning the role of solvent with respect to surface area. Typical data for these catalysts are shown in Table 1 and, based on this, the specific activities can be calculated. For reaction temperatures 450–470°C aqueous preparations ($3\ m^{-2}\ g^{-1}$) give (k_1+k_2)/sa values of ca. 400 g $m^{-2}\ h^{-1}$, whereas non-aqueous preparations ($8–23\ m^2\ g^{-1}$) give corresponding values of ca. 300 g $m^{-2}\ h^{-1}$. It is therefore considered that within the experimental accuracy of the data, the specific activities of these catalysts are broadly similar and the primary reason for the

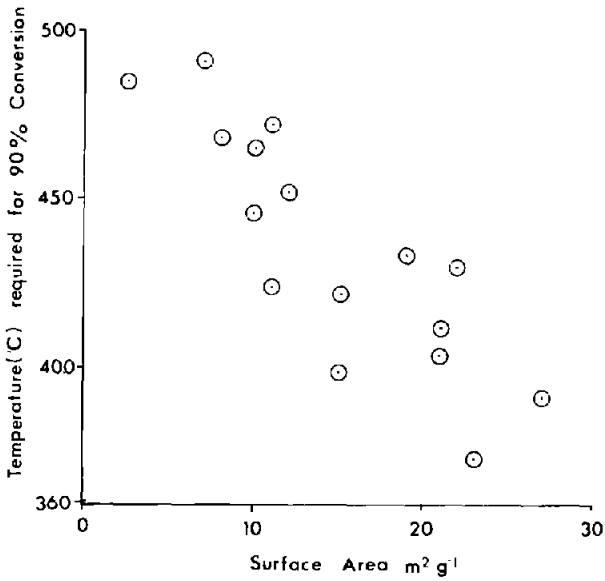


Fig. 1. Correlation between catalyst surface area and activity. GHSV = 1000 h^{-1} [25,26].

TABLE 1

Comparison of catalytic performance of non-promoted catalysts prepared by standard HCl method: V : P = 1 : 1.2, 1.5 mol-% n-butane in air

Ref.	Solvent	Catalytic performance					Final surface area ($m^2 g^{-1}$)	$\frac{k_1+k_2}{sa}$ ($g\ m^{-2}\ h^{-1}$)
		T (°C)	GHSV (h^{-1})	Conv (mol-%)	Sel (mol-%)	Yield (mol-%)		
24	H_2O	453	578	90	53.2	47.9	3.2	415
25,26	THF	420	1000	90	63.1	56.8	22	105
	isobutanol	374	1000	90	69.0	62.1	23	100
	methanol	468	1000	90	59.1	53.2	8	288

enhanced activity of the non-aqueous preparation is that it can reliably give higher surface area catalysts. The importance of increasing surface area for this catalyst is demonstrated by Fig. 1. Attainment of high conversion at lower temperatures improves maleic anhydride selectivity [37] and is therefore economically beneficial; to achieve such yields a high surface area is required. It is therefore necessary to consider the role of the solvent used to prepare the precursor in the attainment of the final catalyst surface area. The DTA of the aqueous preparation, shown in Fig. 2, is characterised by two distinct endotherms. Harrison [24] initially considered that the lower temperature endo-

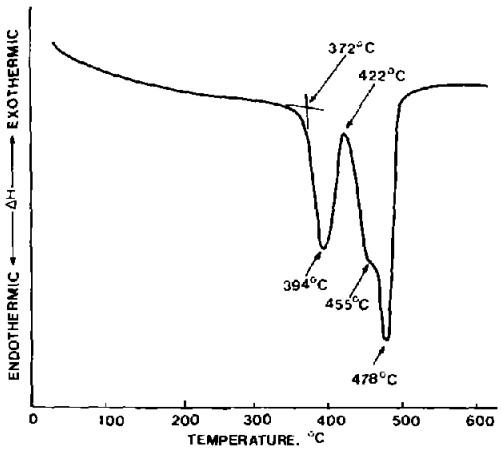


Fig. 2. DTA of catalyst precursor prepared by standard aqueous hydrochloric acid digestion procedure, V : P = 1 : 1.2 [24].

therm was due to dehydration and the higher one to dehydration, coupled with the phase change of precursor (now known to be $VOHPO_4 \cdot 0.5H_2O$) to final catalyst ($(VO)_2P_2O_7$). Thermogravimetric analysis (TGA) measurements gave mass losses which agreed with this conclusion [24]. Consequently they recommended multistep pretreatments to obtain the optimum surface area. Experiments have shown that the DTA/TGA of precursors can show variation in the intensities of these endotherms and in extreme cases may only give a single endotherm [16,24]. Literature data [25] show that when THF is used as a solvent the proportion of the active $(VO)_2P_2O_7$ in the final catalysts is decreased slightly by the addition of water. In addition, the maximum catalyst content of $(VO)_2P_2O_7$ observed is around 90% by mass [25] and consequently about 10% by mass of the catalyst is structurally unaccounted for. Studies [38–41] have shown that the hydrochloric acid digestion method gives rise to $VO(H_2PO_4)_2$ in substantial quantities in addition to the main precursor phase $VOHPO_4$ whether aqueous or non-aqueous solvents are utilised, and it has also been shown [38] that the DTA endotherm in the region 350–400°C was due to the decomposition of this compound. $VO(H_2PO_4)_2$ has been subsequently found to decompose at about 360°C to give $VO(PO_3)_2$ [20] which is a less active catalytic phase than $(VO)_2P_2O_7$ [19]. It is considered that the decomposition of $VO(H_2PO_4)_2$, which occurs at a lower temperature than the transition of $VOHPO_4$ to $(VO)_2P_2O_7$, impedes the attainment of a high surface area catalyst. Removal of $VO(H_2PO_4)_2$ by a simple solvent extraction method [38] results in a final catalyst with an increased surface area. Other workers [42] have subsequently shown that solvent extraction of the precursor can be beneficial. It is therefore concluded that the standard hydrochloric acid digestion preparation procedure prepares $VOHPO_4$ and $VO(H_2PO_4)_2$ as precursor

compounds (in addition, it is probable that other minor VPO compounds are also prepared) and use of a non-aqueous solvent decreases the formation of $VO(H_2PO_4)_2$ relative to $VOHPO_4$. This is consistent with the observation that most patent preparations use V:P atomic ratios of 1:1.15 or 1:1.20, since $VOHPO_4$ has a V:P ratio of unity, the excess phosphorus added in the preparation makes the deleterious compound $VO(H_2PO_4)_2$ with V:P=2. However, excess phosphorus above V:P=1:1 is required to maximise the yields of $(VO)_2P_2O_7$ [25] and a possible reason for this will be discussed subsequently. Mount and Raffelson [33] have also shown that catalyst precursors comprising of $VOHPO_4$ and $VO(H_2PO_4)_2$ can be prepared in the absence of hydrochloric acid, or other reducing agents, by heating V_2O_5 and H_3PO_4 with H_3PO_3 at ca. 150°C in an autoclave. Hence, promotional effects for these catalysts will be considered, together with those for catalysts prepared by the hydrochloric acid digestion method.

Although most literature deals with the promotion of activity/selectivity for catalysts containing mainly $(VO)_2P_2O_7$, the effect of promoters on a number of other compounds have also been studied, e.g. α-$VOPO_4$+$(VO)_2P_2O_7$, β-$VOPO_4$, $VO(PO_3)_2$, however, these studies only account for a minor part of the total literature concerning promoted catalysts.

Moser and Schrader [43] have prepared pure $(VO)_2P_2O_7$ and used this as a model compound for catalytic studies. Careful examination of their X-ray diffraction and IR data with that previously published for the active patented B phase prepared by the hydrochloric acid digestion method [25,38] indicates that, as stated by a number of workers [2,4], the data are similar. However, some minor differences are apparent, in particular the reflection at d=5.7 Å present for pure $(VO)_2P_2O_7$ is not found for the active B phase [25]. In addition, it is interesting to note that the pure $(VO)_2P_2O_7$ exhibited significantly different catalytic behaviour than the B phase. Using pure $(VO)_2P_2O_7$ Moser and Schrader found that the partial oxidation products 1-butene, 2-butene and 1,3-butadiene were present at selectivities of up to 5%, but with the B phase, the only reported carbon containing by-products are carbon monoxide and carbon dioxide, and, significantly, hydrocarbon by-products are not observed. In addition, the activities and maleic anhydride selectivities observed for pure $(VO)_2P_2O_7$ are considerably lower than those reported for B phase containing catalysts [25,38]. These observations indicate that some significant differences may exist between the active B phase and the pure $(VO)_2P_2O_7$. Although a detailed surface and solid-state study is required to resolve this question, it is possible that the active B phase is not a stoichiometric V:P=1:1 oxide phase, but contains small amounts of additional phosphorus within the lattice that is insufficient to markedly change the powder X-ray diffraction pattern. This would then be a possible explanation for the observation that the highest content of B phase in catalysts is obtained for preparations with V:P ratios of 1:1.1–1:1.2 [25] and not 1:1. The presence of the excess phosphorus is known

to make these catalysts more resistant to oxidation of the bulk solid [28]. In addition, two similar VPO phases have been observed in the literature and are as yet to be identified. They are denoted as B′ [38] and β^* [44,45] and are identified by distinct X-ray powder diffraction patterns. Evidence indicates that the B phase and the B′ phase are related by a redox mechanism [38] and it is considered that the phases $(VO)_2P_2O_7$, B, B′ and β^* form a structurally related series of phases, with slight variations in V oxidation state and V:P ratio. Although there are potential differences in the B phase and $(VO)_2P_2O_7$ for the purposes of this review, they will be collectively referred to as $(VO)_2P_2O_7$.

4. PROMOTIONAL EFFECTS FOR VANADIUM PHOSPHORUS OXIDE CATALYSTS

The preceding discussion indicates that most of the catalysts described in the literature contain specific VPO compounds which are defined mainly by the method of preparation. In general the methods of preparing promoted catalysts are the same as those discussed for the unpromoted catalysts. Promoter compounds can be added either (a) together with the vanadium and phosphorus compounds prior to preparation of the catalyst precursor, or (b) by impregnation of the catalyst precursor prior to formation of the final catalyst by heat treatment. In this review the effects of the promoters therefore can be discussed with respect to the following distinct VPO compounds known to be present in catalysts: (a) $(VO)_2P_2O_7$, and $(VO)_2P_2O_7+\alpha VOPO_4$ and (b) other compounds, e.g. β-$VOPO_4$ and $VO(PO_3)_2$.

4.1. Effect of promoters on $(VO)_2P_2O_7$

The catalytic performance of promoted VPO catalysts containing mainly $(VO)_2P_2O_7$ (or B phase) is summarized in Tables 2, 3 and 4. Tables 2 and 3 give the data for catalysts prepared using hydrochloric acid as a reductant with aqueous and non-aqueous solvents respectively. Table 4 gives the data for catalysts containing $(VO)_2P_2O_7$ prepared using other reductants. The catalysts can be broadly divided into two categories: (a) Type 1 promoted formulations, those containing P:V atomic ratios > 1 with correspondingly high promoter:V atomic ratios, i.e. excess P:promoter ratio is approximately unity, and (b) Type 2 promoted formulations, containing low promoter:V atomic ratios. Type 1 promoted catalysts were the first formulations to be disclosed, but recently there has been increased interest in the type 2 formulations.

4.1.1. Type 1-promoted catalysts

Promoters used in this type of catalyst are typically Co, Cd, Ni [60]; Zn, Bi, Cu, Li [46–48]; U [49–55]; Zr [56–58]; and Mg [59]. This type of formulation has been disclosed for aqueous and non-aqueous hydrochloric acid digestion

TABLE 2

Catalytic performance of promoted vanadium phosphorus catalysts prepared using standard aqueous HCl method

No.	Ref.	Company	Catalyst formulation	Pretreatment	Catalytic performance						Comments
					n-Butane (%)	T (°C)	GHSV (h^{-1})	Conv. (mol-%)	Sel. (mol-%)	Yield (mol-%)	
1	24	Chevron	V:P=1:1.2	Standard[a]	1.5	525	700	90	41	37	Surface area of final catalysts ca. 3 m^2 g^{-1}
			V:P=1:1.2	Multistep[b]	1.5	465	813	90	57.4	51.6	
	26	Chevron	V:P=1:1.2	Multistep	1.5	485	1000	90	50.6	45.6	
			V:P=1:1.2	Multistep	1.5	425	500	34	59	20	
			V:P=1:1.2	Multistep	1.5	385	500	13	69	9	
2	46	Standard Oil Co., Indiana	V:P=1:1.15	Standard	1:1.4	450	3600	-	-	30.8	No multistep pretreatment
			V:P:Zn=1:1.15:0.04		1:1.4	450	3600	-	-	30.2	V:P:X=1:1.15:0.16-0.2
			V:P:Zn=1:1.15:0.08		1:1.4	450	3600	-	-	44.8	X=Fe,Mn,Sn,Cr give low MA yields (<14%)
			V:P:Zn=1:1.15:0.19		1:1.4	450	3600	-	-	50.9	
			V:P:Bi=1:1.15:0.19		1:1.4	450	3600	-	-	37.3	
			V:P:Cu=1:1.15:0.19		1:1.4	450	3600	-	-	45.6	
			V:P:Li=1:1.15:0.19		1:1.4	450	3600	-	-	36.7	
3	60	UCB	V:P=1:1.14	Standard	1.5	450	1893	-	-	29.6	Specifically claims multistep pretreatment not required.
			V:P:Co=1:1.14:0.19		1.5	450	1893	-	-	54.7	
			V:P:Co=1:1.14:0.10		1.5	480	1893	-	-	45.0	
			V:P:Cd=1:1.14:0.10		1.5	480	1893	-	-	46.5	
			V:P:Cd=1:1.14:0.05		1.5	490	1893	-	-	50.4	
			V:P:Ni=1:1.14:0.19		1.5	440	1893	-	-	52.1	
			V:P:Co=1:1.1:0.25		1.5	453	1893	-	-	43.0	
			V:P:Co=1:1.14:0.19		1.5	431	1439	-	-	46.0	
4	49	Standard Oil, Ohio	V:P:U=1:1.15:0.20	Standard	1.5	482	2667	-	-	50.0	No multistep pretreatment.
5	56	Mobil	V:P=1:1.2	Standard	3.8	460	312	56	53.6	30.0	$Zr_2P_2O_7$ present. Fluidised bed reactor. No multi step pretreatment. Similar catalysts in ref. 112.
			V:P:Zr=1:1.2:0.13		3.8	400	312	80	61.3	49.0	
			V:P:Zr=1:1.2:0.13		3.8	425	468	77	54.5	42.0	
			V:P:Zr=1:1.2:0.13		3.8	375	94	93	58.1	54.0	

TABLE 2 (continued)

No.	Ref.	Company	Catalyst formulation	Pretreatment	Catalytic performance						Comments
					n-Butane (%)	T (°C)	GHSV (h^{-1})	Conv. (mol-%)	Sel. (mol-%)	Yield (mol %)	
6	59	Monsanto	V P = 1.1.05	Standard	3.0	430	2550	-	-	23.3	No multistep pretreatment.
			V P Mg = 1:1 05 0.05		3.0	423	2550	-	-	22.7	
			V P Mg = 1 1.1.0.05		3.0	446	1500	-	-	42.5	
			V:P·Mg = 1 1.05·0.0125		3.0	462	2630	-	-	25.7	
			V P Mg = 1 1.1 0.1		3.0	442	1460	-	-	38.5	
			V P Mg = 1 1.1 0.1		3.0	433	1550	-	-	34.4	
7	70	Mobil	V P = 1 1.06	Standard	1.0	490	866	-	-	24.3	Low levels of promoters At low levels Co,Cu,U,Ni,Tl all poor. Low levels of Cr,Hf,La,Zr,Ce also beneficial.
			V P Hf = 1·1.06 0.03		1.0	490	866	-	-	42.0	
			V.P:Fe = 1 1.06 0.031		2.0	470	874	-	-	45.0	
			V·P.Fe = 1 1.06:0.056		2 0	455	874	-	-	37.9	
			V P Fe = 1:1 06 0.014		2.0	530	874	-	-	36.1	
8	73	Monsanto	V P = 1 1.05	Standard	3.14	494	2480	-	-	21.0	Ti permits lower T operation.
			V.P·Ti = 1:1.05 0.07		2.5	438	2500	-	-	20.5	Synergistic effect with Ti and Li.
			V P Li = 1 1.05:0.10		3.0	502	2500	-	-	21.3	Similar effect with Ti and Si, see ref.
			V P Ti Li = 1 1.05.0.04 0.1		2.95	448	2510	-	-	30.6	133 No multistep pretreatment.
			V.P.Ti Li = 1 1.05 0.07·0.05		2.75	414	2460	-	-	23 3	
			V P Ti:Li = 1.1.05 0.07 0.1		3.05	408	2550	-	-	17.1	
			V P Ti Li = 1 1.05.0.04·0.2		3.0	429	2510	-	-	28.3	
			V P Ti Li = 1 1.05 0.07·0.2		3.0	420	2500	-	-	28.6	
9	74	Halcon	V P = 1:1.15	Standard	1.5	500	1200	80	47	37 6	Zn, B, Sb and additional P added by post impregnation. Synergistic effect with Ti,Zn,B,Sb and additional P.
			V.P Zn = 1 1.15 0.03		1.5	455	1200	80	55	44.0	
			V P Ti = 1 1.15 0.06		1 5	460	1200	80	47	37.6	
			V P Ti.Zn = 1 1.15 0.04 0.04		1.5	460	1200	80	53	42.4	
			V.P Ti Zn = 1·1.15.0.06 0.04		1.5	420	1200	80	58	46.4	
			V P.Ti Zn.B = 1 1.15 0.06 0.04 0.05		1 5	415	1200	80	55	44.0	
			V P Ti Zn Sb = 1·1.15.0.06 0.04 0.02		1.5	415	1200	80	61	48.8	
			V P Ti Zn P = 1 1.15·0.06 0.04 0.05		1.5	420	1200	80	62	49.6	

10	74	Halcon	(i) V P Ti Cu = 1.1.15 0.06 0.04	Standard	1.5	425	1200	80	50	40.0	(i) promoters added by impregnation
			(ii) V P:Ti·Cu = 1.1.15.0.06 0.04		1.5	455	1200	80	45	36.0	(ii) promoters added in bulk
11	38	ICI	V P = 1 1.2	Standard	1.5	420	2000	72	60	43.2	promoters added by impregnation prior
			V.P Co = 1 1.2 0.75		1.5	420	1600	87	68	58.9	to heat treatment.
	66		V·P La = 1 1.2 0.27		1.5	420	2500	88	68	59.8	
	91		V·P Ce = 1.1.2.0.125		1.5	420	1000	95	53	50.4	
12	65	Petrotex	V P = 1:1 12	Calcined	1.27	380	850	82	52	42.6	Synergistic effect of Si and B with Zn
			V.P Zn.Si = 1 1.15 0.19·0.17	350°C	1.10	388	1346	80	73	58.4	and U.
			V:P U = 1 1.1:0.25		1.03	360	1346	83	43	35.7	
			V.P.U B = 1·1.15 0.025:0.01		1.16	360	1347	81	50	40.5	
13	39	Standard	(a) V P.Mo = 1·1.28.0.03	Multistep	1.05	413	1200	88	63	55.4	Solvents added to aid precipitation.
		Oil Co.,	(b) V P Mo = 1 1.28:0.03		1.05	444	1200	80	63	50.4	Catalysts contain $VOHPO_4$ and
		Indiana	(c) V P·Mo = 1 1.28.0.03		1.05	434	1200	86	66	56.8	$VO(H_2PO_4)_2$.
			(d) V·P·Mo = 1:1.20 0.03		1.05	417	1200	87	67	58.2	(a) 72% $VOHPO_4$; 16% $VO(H_2PO_4)_2$
											(b) 63% $VOHPO_4$; 13% $VO(H_2PO_4)_2$
											(c) 81% $VOHPO_4$; 0% $VO(H_2PO_4)_2$
											(d) 85% $VOHPO_4$; 0% $VO(H_2PO_4)_2$
14	39	Standard	V P Zn = 1:1.20:0.18	Multistep	1.05	442	1200	77	63	48.5	Methanol added to reaction mixture
	40	Oil Co.,	V:P W = 1.1.20.0.03		1.05	436	1200	82	61	50.0	prior to solvent extraction.
		Indiana									
15	40	Standard	(a) V P.Mo = 1 1.28 0.03	Multistep	1.05	441	1200	81	65	52.7	(a) MeOH only added.
		Oil Co.,	(b) V:P:Mo = 1 1.28 0.03		1.05	432	1200	86	66	56.8	(b) MeOH + phthalic anhydride added.

[a]Standard procedure: heating to operating temperature in n butane/air mixture.
[b]Multistep procedure: initial heating to 380°C followed by heating to 480°C in n-butane/air mixtures.

TABLE 3

Catalytic performance of promoted vanadium phosphorus catalysts prepared using standard non aqueous HCl method

No.	Ref.	Company	Catalyst formulation	Solvent	Pre-treatment	Catalytic performance						Final surface area (m^2/g)	$\frac{k_1+k_2}{sa}$ ($g\ m^{-2}\ h^{-1}$)	Comments
						n Butane (%)	T (°C)	GHSV (h^{-1})	Conv. (mol %)	Sel (mol-%)	Yield (mol %)			
1	26	Chevron	V:P = 1:1.2	THF	Multistep	1.5	420	1000	90	63.1	56.8	22	105	
			V:P = 1:1.2	isobutanol	380 °C and 480 °C in n-butane/air	1.5	374	1000	90	69.0	62.1	23	100	
	81	ICI	V:P = 1:1.2	isobutanol	Standard	1.5	385	750	74	67.4	50.0	14	72	
			V:P = 1:1.2	isobutanol	Standard	1.5	340	205	65	80.0	52.0	14	15	
2	60	UCB	V:P:Co = 1:1.14:0.19	isobutanol	Standard	1.5	387	1028	90	65.7	59.2	20*	118	*sa obtained from ref. 66
			V:P:Co = 1:1.14:0.19	methanol	Standard	1.5	405	1028	-	-	60.0	-	-	
3	50	Standard Oil Co., Indiana	V:P:U = 1:1.28:0.10	isobutanol + benzene	Standard	1.05	406	1000 – 2400	75	67.0	50.3		-	
4	47	Standard Oil Co., Indiana	V:P:Zn = 1:1.28:0.10	methanol + benzene	Standard	1.05	427	1000 – 2400	86	53.0	45.6	-	-	
5	81	ICI	V:P:Mo = 1:1.2:0.04	isobutanol	Standard	1.5	340	300	85	82.0	70	24	24	Low levels of promoters
			V:P:Mo = 1:1.2:0.026	isobutanol	Standard	1.5	385	726	90	66.7	60	22	76	
			V:P:Mo = 1:1.2:0.04	isobutanol	Standard	1.5	385	750	98	67.3	66	24	122	
			V:P:Mo = 1:1.2:0.076	isobutanol	Standard	1.5	385	416	95	68.4	65	16	78	
			V:P:Nb = 1:1.2:0.07	isobutanol	Standard	1.5	340	100	96	73.0	70	15	21	
6	82	Standard Oil Indiana	(a) V:P:Mo = 1:1.3:0.03	methanol	Standard	1.05	436	1200 – 1300	81.1	65.8	53.4	-	-	P_2O_5 as P source.
			(b) V:P:Mo = 1:1.3:0.03	methanol	Standard	1.05	436	1200 – 1300	86.0	62.1	53.4	-	-	(a) P_2O_5 added initially
			(a) V:P:Zn = 1:1.2:0.18	methanol	Standard	1.05	404	1200	82.0	62.0	50.8	-	-	(b) 80% P_2O_5 added initially, 20%
			(b) V:P:Zn = 1:1.28:0.10	methanol	Standard	1.05	404	1200	88.0	55.0	48.4	-	-	added after HCl digestion

7	81	Standard Oil Co., Indiana	V:P=1:1.16 (a)	isobutanol	Standard	1.64	418	1800	100	61	61.1	-	-	Catalysts supported on Al_2O_3.
			V:P=1:1.16 (b)			1.64	456	1714	89	49	43.5	-	-	
			V:P:U=1:1.20:0.02 (a)											
			V:P:U=1:1.20:0.2 (b)			1.69	441	1659	94	67	62.5	-	-	(a) 45% Al_2O_3.
						1.59	401	1800	89	63	56.4	-	-	(b) no Al_2O_3.
8	41	Standard Oil Co., Indiana	V:P:Mo=1:1.32:0.014	methanol/ σ-xylene	Standard	(a) 1.06	443	1200	75	62	47.0	-	-	Similar to ref. 82 with H_3PO_4 as P source. (a) 13 days. (b) 27 days. (c) 40 days. Catalysts precursor comprised 70% $VOHPO_4$ and 13% $VO(H_2PO_4)_2$.
						(b) 1.08	443	1200	87	61	52.8	-	-	
						(c) 1.08	443	1200	90	64	57.2	-	-	
9	129	Monsanto	V:P:Fe:Li=1:1.2:0.0016:0.003	isobutanol	Multistep	1.9	413	1150	78	70	54.5	-	-	1.8% H_2O added during pretreatment.
	130		V:P:Li=1:1.2:0.003			1.9	413	1150	69	66	45.6	-	-	
10	83	Amoco	V:P:Mo=1:1.28:0.13		Standard	1.1	-	1200	89	69	61.4	40	66.2	1% H_2O added to reactants. 53 days lifetime.

TABLE 4

Catalytic performance of promoted vanadium phosphorus catalysts prepared using HCl alternatives

No.	Ref.	Company	Catalyst formulation	Reductant/ Solvent	n Butane (%)	T (°C)	GHSV (h^{-1})	Conv. (mol %)	Sel. (mol-%)	Yield (mol %)	Comments
1	61,62, 63,64	Petrotex/ Denka	V P Cu Te Li = 1 1.2 0.026 0.026 0.039	oxalic acid/ isopropanol	1.5	483	700 -1800	83.6	42.1	35.2	Catalysts supported on Nb_2O_5
			V P Cu Ce Li = 1 1.2 0.026 0.026 0.09		1.5	445	"	59.4	49.1	29.2	*CrO_3 0.4-2%
			V P Cu Cr Te Li = 1 1.2 0.026 0.026 0.052 0.039		1.5	450	"	84.8	52.2	44.2	promoter.
			V P Cu Te Li = 1 1.14 0.034 0.011 0.045		1.5	447	"	88.5	54.8	48.5	
			V P Cu Te Li* = 1 0.9 0.023 0.023 0.068		1.5	457	"	68	22.4	15.2	
			V P Cu Te Li = 1 1.1 0.016 0.016 0.022		1.5	444	"	84	57.9	48.6	
2	48	Standard Oil Co., Indiana	V P Cr = 1 1.2 0.2	hydrazine hydrate/water	1.5	523	3000	75.4	18.3	13.8	
			V P Cu = 1 1.2 0.2		1.5	525	3000	92.6	25.7	23.8	
			V P Zn = 1 1.2 0.2		1.1	462	2000	82.3	22.0	18.1	
3	51	Standard Oil Co., Indiana	V P U (W) = 1 1.15 0.20 (0.166)	tungsten metal/ water	1.6	513	2000 -3000	100	33.2	33.2	Catalysts prepared by ambient reflux
					1.2	456	"	99.8	46.7	46.6	
					1.5	456	"	100	48.6	48.6	
			V P U = 1 1.15 0.20	hydrazine hydrate/water	1.5	512	"	97.6	33.4	32.6	
					1.2	486	"	56.9	48.0	27.3	
			V P U = 1 1.15 0.20	oxalic acid/ water	1.1	537	"	11.8	11.0	1.3	
			V P U = 1 1.15 0.20	HCl/water	1.4	502	"	92.6	42.0	38.9	
					1.3	468	"	75.3	50.3	37.9	
4	52	Standard Oil Co., Indiana	V P U (W) = 1 1.15 0.20 (0.166), 135 °C	tungsten metal/ water	1.1	477	2000 -3000	99.1	45.7	45.3	Catalysts prepared by autoclaving at 135-180 °C.
			V P (W) = 1 1.15, (0.166), 135 °C		1.2	534	"	98.1	37.3	36.6	Additional
			V P U (W) = 1 1.15 0.20 (0.166), 180 °C		1.0	506	"	98.6	36.6	36.1	promoters, e.g.
			V P U (W) = 1 1.15 0.20 (0.166), 180 °C		1.1	466	"	94.0	45.3	42.6	Co,Cu,Bi,La
			V P U Co (W) = 1 1.15 0.20 0.05 (0.166), 180 °C		1.6	495	"	88.4	27.6	24.4	deleterious.
			V P U Cu (W) = 1 1.15 0.20 0.05 (0.166), 180 °C		1.8	509	"	100	25.0	25.0	Additional W no
			V P U Bi (W) = 1 1.15 0.20 0.05 (0.166), 180 °C		1.4	540	"	97.4	26.5	25.8	significant effect.
			V P U La (W) = 1 1.15 0.20 0.02 (0.166), 180 °C		1.5	532	"	100	24.3	24.3	
			V P U Bi (W) = 1 1.15 0.20 0.05 (0.166), 180 °C		1.5	478	"	92.4	43.6	40.3	

5	69	Mitsubishi	V.P=1 1	hydrazine	1.5	418	500	95.0	44.0	41.8	(a) Promoters in
			(a) V.Fe.P=0.95:0.05 1	hydrate/water	1.5	413	500	97.0	45.6	44.2	solid solution
			(a) V Fe P=0.9 0.1 1		1.5	380	500	95.4	47.1	44.9	$(VO)_2P_2O_7$
			(b) V·Fe·P=0.9:0.1:1		1.5	370	500	90.4	41.6	37.6	(b) Fe not in solid
			(a) V Cr P=0.9 0.1 1		1.5	365	500	94.0	44.1	41.4	solution
			(a) V Fe Cr P=0.94·0.04·0.02·1		1.5	364	500	95.4	43.7	42.2	
6	71	Monsanto	V·P=1 1.06	H_3PO_3/water	1.9	413	1540	-	-	28.9	V_2O_5, H_3PO_3 and
	72		V.P.B=1.1.06:0.05		1.9	363	1540	-	-	28.9	H_3BO_3 heated in au-
											toclave at 155°C
7	55	Standard	V:P U=1:1.2 0.2	hexachloro	1.6	431	2120	63.6	57.6	36.6	
		Oil Co.,	V.P Co=1.1.2.0.2	butadiene	1.1	437	1890	89.3	55.6	49.7	
		Ohio									
8	131	-	V·P·Ti· 1:1.1·0.2	H_3PO_3/	1.5	455	2290	85.0	57.5	48.9	H_3PO_3 added to give
			V.P:Ti:Ni=1 1.1·0.2.0.02	isobutanol	1.5	460	2290	87.2	57.5	50.1	V oxidation state 4.3.
			V·P·Ti·Ni:Li=1 1.1 0.2 0.03·0.01		1.5	460	2290	81.9	58.7	48.1	
			V P:Ti:Fe.Mg=1:1.1:0.2.0.02:0.005		1.5	460	2290	91.4	52.9	48.4	
9	57	Mobil	V P=1.1.2	ethylene glycol	3.8	450	624	54.7	60.0	32.8	
	58		V.P Zr=1 1.2 0.13	ethylene glycol	3.8	410	624	63.2	61.0	38.5	
			V P Zr=1 1.2·0.09	formaldehyde	3.8	425	468	61.1	61.0	37.3	
			V:P Zr=1:1.2 0.13	sucrose	3.8	410	624	62.8	62.0	38.9	
10	78	Du Pont	V:P In=1:1.28:0.03	benyzl alcohol/	1.5	420	900	-	-	37.2	Catalysts contain 2%
	79			isobutanol			-1800				SiO_2.
			V P Ta=1 1.28 0.02		1.5	420	"	-	-	32.0	
			V P In:Sb=1·1.28 0.05 0.05		1.5	420	"	-	-	41.4	
11	80	Monsanto	V P=1:1.01	SO_2	1.5	360	1450	82.0	73.8	60.5	
			V P:Zn=1:1.0:0.05		1.5	360	1450	80.0	66.2	53.0	

preparations as well as with alternative reductants to hydrochloric acid. The V : P atomic ratio is typically in the range 1 : 1.15 to 1 : 1.20 and the V : promoter ratios are correspondingly 1 : 0.15 to 1 : 0.20 for optimum performance. In addition, the promoters are added to the preparation procedure prior to formation of the precursor. Boghosian [46] initially disclosed a formulation with Zn, Bi, Cu and Li (line 2, Table 2), and showed that these promoters increased the maleic anhydride yield, whereas Fe, Mn, Sn and Cr at these high levels had a deleterious effect. Subsequently Lemal and Vekemans [60] showed similar beneficial effects for Co, Cd and Ni (line 3, Table 2, line 2, Table 3). These authors specifically stated that the multistep pretreatment procedure [24], during which the temperature of pretreatment is sequentially raised, to convert the precursor to the final catalyst was not necessary with these catalysts. This is a central feature of these catalysts and similar results were obtained for U (line 4, Table 2, line 3, Table 3, lines 3, 4, 7, Table 4), Zr (line 5, Table 2, line 9, Table 4), Mg (line 6, Table 2) and Ti (line 8, Table 4). Kerr [61] and Kerr and Barone [62–64] (line 1, Table 4) have shown for a basic formulation V : P : Cu, additionally promoted by Te or Ce and Li, that when V : P = 1 : 1.2 and total promoters: V < 0.1, then the resultant yield of maleic anhydride is 30–35 mol-% at 60–85% conversion. When V : P = 1 : 1.1 for the same promoter concentrations then the yield is increased to 48–51 mol-% at 90% conversion; but when V : P = 1 : 0.9 or 1 : 1 for these promoters then the yield is reduced to 2–15 mol-% and the major products are carbon monoxide and carbon dioxide, indicating that excess phosphorus must be present to gain beneficial effects. This effect was also shown by Bremer [52] (line 4, Table 4) for a V : P : U catalyst when additional promoters, e.g. Co, Cu, Bi and Zn, gave reduced maleic anhydride yields. Studies have also shown (line 6, Table 2) that with a catalyst formulation V : P : Mg the effect of the promoter was not significant unless P excess > Mg. These results may indicate that for these B phase containing catalysts the optimum P : V ratio may be in excess of unity, i.e. above the stoichiometric requirement for pure $(VO)_2P_2O_7$.

Barone and Kerr [65] (line 12, Table 2) have shown that the performance of the Zn and U promoted formulations can be further improved by addition of Si and B to the catalyst, the effect being mainly due to an improvement in selectivity. Additionally, Dolhyj and Pepera [54] (line 7, Table 3) have shown that the U promoted formulations can be successfully operated when supported on Al_2O_3. Use of the support improves the overall yield of maleic anhydride, whilst decreasing the amount of active catalytic component utilised and therefore this approach would appear to warrant a further detailed study.

However, the striking feature of all these catalysts is that the optimum yields obtained are only the same as those obtained from equivalent non-promoted catalysts, but prepared using a detailed heat treatment procedure [24,26]. Surface area data for the final catalysts is not readily obtainable, but data is available for the V : P : Co = 1 : 1.2 : 0.2 formulation prepared using isobutanol as solvent [60,66]. Using these data the specific activity of this catalyst is determined

to be $(k_1+k_2)/sa=118$ g m^{-2} h^{-1} at 387°C and GHSV = 1028 h^{-1}. The specific activity for a non-promoted catalyst at the same condition can be calculated to be $(k_1+k_2)/sa=100$ g m^{-2} h^{-1} [25]. Therefore, the promotional effect of the addition of a massive amount of cobalt is not particularly significant. Hence, the role of the promoter for this type of formulation must be considered. It is interesting to note that for the catalyst V:P:Zr = 1:1.2:0.13, $Zr_2P_2O_7$ was present in the final catalyst as determined by X-ray diffraction [56]. Unfortunately no further X-ray data are available for the previously cited catalysts, but Brutovsky and Gerej [67,68] have studied a range of promoted catalysts with similar formulations and preparation procedure. They found that a number of cations, e.g. Co, Fe, Li, Zn, Ce, Mn, Sn, Lu, formed solid solutions in $(VO)_2P_2O_7$. The formation of solid solutions has been confirmed the by studies of Otake [69] (line 5, Table 4) where solid solutions of $(VO)_2P_2O_7$ with Fe and Cr have been shown to be active catalysts. However, it is important to note that the levels of Cr and Fe utilised are much lower than those used in the previous study with these promoters [46]. It is therefore probable that the promoters in previously cited catalysts are present in solid solution with $(VO)_2P_2O_7$. Since these catalysts do not require the multistep pretreatment, which is required for the unpromoted catalysts due to the presence of $VO(H_2PO_4)_2$, it is proposed that the role of these promoters is to form solid solutions of the type $((VO)_{1-x}M_x)_2P_2O_7$. In this way it is further proposed that the excess phosphorus is prevented from forming $VO(H_2PO_4)_2$, or other unwanted compounds, thereby ensuring that a high surface area catalyst is reliably obtained without recourse to any detailed heat treatment procedure. Hence, it is the view of this author that these promoters act as phosphorus scavengers by either formation of metal phosphates or by formation of solid solutions. When solid solution of the type $((VO)_{1-x}M_x)_2P_2O_7$ is formed, this ensures that the active catalyst compound contains the excess phosphorus required to maintain its stability in an oxidising environment. However, these solid solutions may have, in certain cases, higher specific activities and this aspect will be dealt with in a subsequent section.

4.1.2. Type 2-promoted catalysts

Recently a number of formulations have been disclosed in which the promoter:V atomic ratios are very low. Mobil researchers [70] (line 7, Table 2) have investigated a range of promoted catalysts with V:P = 1:1.06 and containing low concentrations of promoters. The level of promoter added is considered to be too low to be having the structural effect discussed for type 1 promoted catalysts. This is confirmed since low levels of Co, Cu, U and Ni were all found to give decreased maleic anhydride yields, whereas in type 1 catalysts at higher concentrations these promoters all gave increased yields [46,49,60]. The best effects were obtained with Cr, Fe, Hf, Zr and Ce and the data presented for iron showed that the optimum V:Fe atomic ratio was 0.03. This

effect is similar to that found by Otake [69] (line 5, Table 4) for catalysts containing solid solutions of Fe or Cr in $(VO)_2P_2O_7$, where it was shown that if the iron is not present in solid solution then the yield of maleic anhydride is decreased. Otake presents some surface area data and it is possible to calculate the effect of iron in solid solutions on the catalyst specific activity. At the same reaction condition (GHSV = 500 h^{-1}, T = 413–418°C) the specific activity for the unpromoted catalyst is $(k_1+k_2)/sa = 123$ g m^{-2} h^{-1}, whereas that of the promoted $(V_{0.95}Fe_{0.05}O)_2P_2O_7$ is $(k_1+k_2)/sa = 188$ g $m^{-2}h^{-1}$. This indicates that the low level of iron is having a significant promotional effect on the catalyst activity.

Studies have shown that the addition of Ti (line 8, Table 2) or B [71,72] (line 6, Table 4) permits lower temperatures to be used. However, an interesting observation is that synergistic promoter effects can be observed when titanium is used [73]. Addition of Li to V:P:Ti catalysts gives increased yields, whereas addition of Li without Ti has no significant effect. Similar synergistic effects have been observed by McDermott [74] (line 9, Table 2) for V:P:Ti catalysts with additional Zn, B, Sb and P. McDermott (line 10, Table 2) has also investigated the role of post deposition of promoters by impregnation of the catalyst precursor. Beneficial effects are indicated for this procedure, i.e. lower operating temperatures giving higher yields. This may originate from the concentration of the promoter on the catalyst surface, since impregnation in this way would be expected to deposit the promoter preferentially on the surface. However, it must be noted that the impregnation procedure could increase the catalyst surface area which would result in increased activity. Since surface area data is not available for these catalysts it is not possible to distinguish between these effects. Other workers [75,76] have shown that addition of boron can be beneficial for catalysts used in fluidised beds, since it reduces catalyst attrition. A recent disclosure by Glukhovskii et al. [77] has shown that addition of very low levels of selenium to vanadium phosphorus catalysts (V:Se ≤ 1:0.0217) decreases the induction period observed with these catalysts. This initial induction period can be several days, and hence, this observation could be of significance for the commercial operation of fixed bed reactors.

Recently a number of catalysts have been disclosed with low levels of promoters and using alternative reductants to the hydrochloric acid digestion method. Bither [78,79] has studied In, Ta and Sb (line 10, Table 4) using benzyl alcohol/isobutanol as the reductant. Wrobleski et al. [80] has shown (line 11, Table 4) that low levels of zinc can be beneficial in catalysts prepared using SO_2 as the reductant which yields catalyst precursor with a P:V ratio of unity. The effect of low levels of zinc in this case is to improve catalyst selectivity and has no marked effect on catalyst activity.

A large number of studies have centred on the use of molybdenum as a promoter at low levels [35,39,40,41,81-83] (lines 13, 15, Table 2, lines 5, 6, 8, 10,

Table 3). Higgins and Hutchings [81] showed that the optimum level of molybdenum, for the conditions tested, was V:Mo = 1:0.04 and at this level increased catalyst specific activity and maleic anhydride yields were observed. The striking feature of this study is that yields of around 70 mol-% can be obtained with these catalysts which is considerably higher than other catalysts disclosed to date. It is therefore apparent that the promotion action of molybdenum in these catalysts is not just to increase specific activity, but reduces oxidation of maleic anhydride at high conversion levels (> 95%), thereby improving selectivity. It is interesting to note that molybdenum is utilised as a promoter in formulations disclosed by Kerr [84] and Barone [85], but since these catalysts are additionally promoted by Cu, Ni, Co, Cr, Ba and lanthanides, it is not possible to delineate any effect to a specific promoter.

Hutchings and Higgins [38] have noted that removal of $VO(H_2PO_4)_2$ from catalyst precursors prepared using the standard hydrochloric acid digestion procedure results in final catalysts that contain appreciable quantities of a high surface area form of α-$VOPO_4$ in addition to the B phase ($(VO)_2P_2O_7$) and a phase denoted B′. These catalysts were found to be more active than those catalysts containing only $(VO)_2P_2O_7$ and a correlation was noted between the specific activity and the α-$VOPO_4$ content. It is interesting to note that some workers have indicated specific procedures for minimising the formation of $VOPO_4$ [86,87], since it has been considered that catalysts comprising of pure α-$VOPO_4$ are too rich in oxygen to be selective catalysts [88]. However, studies using 1-butene oxidation [89] have shown that mixtures of α-$VOPO_4$ and $(VO)_2P_2O_7$ produced the most selective catalysts, indicating appropriate combinations of α-$VOPO_4$ and $(VO)_2P_2O_7$ may be beneficial. In addition Bremer and Dria [90] have shown that improved catalysts can be prepared by partially reducing α-$VOPO_4$, further confirming that mixtures of α-$VOPO_4$ and $(VO)_2P_2O_7$ are beneficial. Bordes and Coutine [19] have shown that α-$VOPO_4$ and $(VO)_2P_2O_7$ are related by a redox mechanism and the availability of such a reaction may be significant in understanding the enhanced catalytic performance of mixtures of α-$VOPO_4$ and $(VO)_2P_2O_7$. The effects of a range of promoters on catalysts containing α-$VOPO_4$ in addition to $(VO)_2P_2O_7$ have been studied [38,66,91] (line 11, Table 2). Addition of trace concentrations of Co, La, Cr, Cu, Ce and Ni were found to increase significantly the specific activity of the catalysts (e.g. at 420°C, V:P = 1:1.2, $k_1 + k_2/sa = 96\ g\ m^{-2}\ h^{-1}$, V:P:Co = 1:1.2:0.075 $k_1 + k_1/sa = 321\ g\ m^{-2}\ h^{-1}$) and also result in improved selectivity for maleic anhydride. The promotional effect of low concentrations of Cr, La and Ce have been previously noted for catalysts containing mainly $(VO)_2P_2O_7$ [70], but for $(VO)_2P_2O_7$ trace levels of Co, Cu and Ni were found to be deleterious. Therefore Co, Cu and Ni may have a specific promotional action on α-$VOPO_4$, whereas Cr, La and Ce may be predominantly affecting the $(VO)_2P_2O_7$ present in the formulation. However, the yields obtained for the promoted mixtures of α-$VOPO_4$ and $(VO)_2P_2O_7$ are much higher than

those of similar formulations containing only $(VO)_2P_2O_7$ [70]. Ball milling of catalyst precursors in the presence of promoters has also been shown to be beneficial [91,92] and this procedure considerably increases the catalyst surface area from 10–18 $m^2 g^{-1}$ to 30–50 $m^2 g^{-1}$. Ball milling has also been used as a central preparative step by Sucio et al. [75,76] to obtain catalyst particles $<0.5 \cdot 10^{-6}$ m for formulations promoted with B and Al.

Recently researchers at Dupont [93–97] have disclosed a process for the manufacture of maleic anhydride which involves a recirculating catalyst bed. For this application $(VO)_2P_2O_7$ by itself was found to be too weak to be utilised without further modification. To overcome this 10% SiO_2 was added to the $(VO)_2P_2O_7$ via a spray drying process using tetraethylorthosilicate as the source of silica. Free passage of both reactants and products can be achieved with this overlayer and selectivities of >90% maleic anhydride have been observed. While the SiO_2 is not acting as a promoter or a support, this preparative procedure demonstrates a novel method of overcoming the inherent structural weakness of oxide catalysts for use in fluidised bed or recirculating catalyst bed reactors.

4.2. Effect of promoter on other VPO catalysts

The influence of promoters on VPO phases/compounds other than $(VO)_2P_2O_7$ (the B phase) has received scant attention. Hodnett and Delmon [98] have studied the effect of cobalt added by a co-precipitation procedure to a catalyst consisting of β-$VOPO_4$ with a trace of the β^* phase. The structures α- and β-$VOPO_4$ are both comprised layers of VO_6 pseudo-octahedra sharing corners with PO_4 tetrahedra, the α form giving ribbons or sheets in the (001) plane and the β form in the (010) plane. Therefore, the cobalt promoted β-$VOPO_4+\beta^*$ catalysts of these studies are analogous to the promoted α-$VOPO_4$ catalysts studied by Hutchings and Higgins [38]. Hodnett and Delmon showed that addition of cobalt up to Co/V=0.05 increased the specific activity, but as the surface area steadily decreased with increasing cobalt content, the highest conversion was observed with the unpromoted catalysts. Selectivity to maleic anhydride showed an optimum level at Co/V in the range 0.01–0.02, and a distinct maximum in catalyst performance was observed with these formulations. These effects are therefore broadly analogous to those observed for α-$VOPO_4$, but for α-$VOPO_4$ the catalyst surface area was usually observed to increase for promoted catalysts and this may be a consequence of the different layer structures of α- and β-$VOPO_4$.

Ti, Ce and Tl promoted $VO(PO_3)_2$ catalysts have been reported for the selective oxidation of butenes or 1,3-butadiene [99–101], but as yet promoted $VO(PO_3)_2$ catalysts have not been studied for n-butane oxidation. Bordes and Courtine [19] have shown that pure $VO(PO_3)_2$ is not readily reoxidisable and consequently is a poor oxidation catalyst for n-butane, but the addition of suit-

able promoters could facilitate the redox process and high selectivity (66 mol-%) at high conversion (97 mol-%) was observed for 1-butene oxidation. Similar catalysts should therefore be investigated for n-butane oxidation.

Harrison [102,103] has disclosed VPO catalysts prepared with co-precipitated silicon which is added to the preparation as an alkyl orthosilicate. The method of preparation uses benzyl alcohol in place of hydrochloric acid and structural information for these catalysts was not given. The addition of silicon at levels of up to 5.3% were studied and optimum performance was found for 1.5% Si. promotion of both catalyst activity and selectivity was observed with these silicon promoted catalysts.

Moorehead [104,105] has studied the use of mordenite and silicilite to obtain a higher intrinsic surface area (>300 m^2 g^{-1}) for a tin promoted vanadium phosphorus catalyst prepared from NH_4VO_3, H_3PO_4, $SnCl_2$ and HCl/C_2H_5OH (V:P:Sn=1:2.3:0.79). When operated at low n-butane conversions (10–20%) these catalysts give high maleic anhydride selectivities (59–74 mol-%). However, due to the high surface area it can be expected that the selectivities will be particularly susceptible to increases in conversion due to consecutive oxidation of maleic anhydride. Whilst higher conversion with n-butane are not exemplified, butene oxidation at 79% conversion gave a maleic anhydride selectivity of only 40 mol-%, and hence, the use of these high area catalysts can be expected to be restricted to low conversion processes.

Prior to the disclosure of the active and selective V:P=1:1.2 catalysts prepared by the hydrochloric acid digestion procedure, many studies concentrated on higher P:V molar ratios. Catalysts with V:P ratios of 1:4.0 promoted with Sn/Sb and supported on TiO_2 were proposed [106] for the oxidation of mixed butane/butene feedstocks, while vanadium molybdenum oxide catalysts promoted by silver phosphate have also been studied [107]. Monsanto researchers [108–111] disclosed phosphorus rich iron promoted catalysts V:P:Fe=1:4.7:1.6 which were additionally promoted by other metals, e.g. Ba, Cr. The yields obtained with these catalysts were low, <30 mol-%, and it is therefore not surprising that subsequent studies published by Monsanto have concentrated on the more active V:P=1:1.05-1:1.10 catalysts.

5. MECHANISMS FOR PROMOTIONAL EFFECTS

From the preceding discussion it is clear that a wide variety of metal cations can improve the catalytic performance of VPO catalysts containing a range of distinct compounds. Of the studies reported the majority do not attempt to elucidate the mechanism of promoter effects, and to date only a few academic studies have attempted this [42,67,68,98]. Using the approach adopted in this review it is concluded that a prime role of promoters is structural modification of the bulk VPO phases. In particular this is observed for $(VO)_2P_2O_7$ containing catalysts prepared using $P/V>1.1$, with Bi, Cd, Co, Cu, Li, Mg, U, Zn and Zr as promoters. The role of these promoters would appear to be to form phos-

phates with the excess phosphorus present in these formulations, thereby preventing the formation of inactive or deleterious VPO compounds, e.g. $VO(H_2PO_4)_2$. Some studies have shown that phosphates are present, e.g. Zn [42] and Zr [56] which provides supporting evidence for this conclusion. In addition, the promoted catalysts specifically do not require the detailed heat pretreatment required for the unpromoted catalysts [60], and this may be related to the formation of phosphates. However, this is probably not the only role of these promoters but other effects are masked by this dominant effect due to the high concentrations of promoters added. A number of promoters, e.g. Ce, Co, Cr, Cu, Fe, Hf, La, Mo, Nb, Ni, Ti and Zr, have been shown to be effective at very low concentrations. A number of studies [67–69] have shown that these promoters form solid solutions with $(VO)_2P_2O_7$ to form compounds of the type $((VO)_xM_{1-x})_2P_2O_7$, where M = promoter. In addition, recent studies [113,114] have shown that $(VO)_2P_2O_7$ can be oxidised to $(VO)^{2+}_{2-x}(VO)^{3+}_x(P_2O_7)^{4-}_{1-x/2}(PO_4)^{3-}_x$ and this process may be involved in the formation of these solid solutions. Otake [69] showed that these solid solutions exhibit significantly enhanced specific activity, e.g. at GHSV = 500 h^{-1}, T = 413–418°C $((VO)_{0.95}Fe_{0.05})_2P_2O_7$ gave $(k_1+k_2)/sa$ = 188 g m^{-2} h^{-1} compared to 123 g m^{-2} h^{-1} for the unpromoted catalyst. When the iron was added

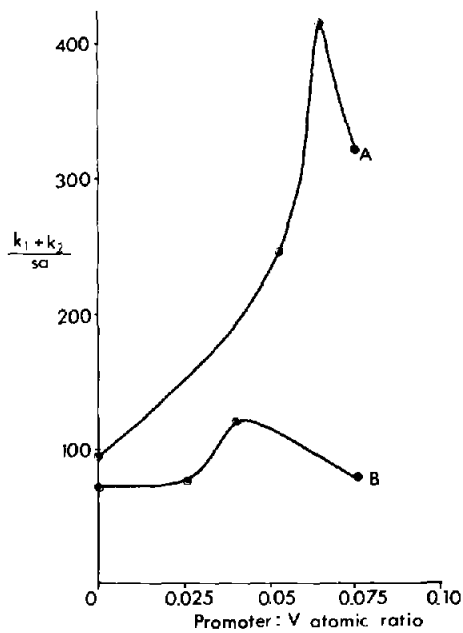


Fig. 3. Effect of promoters on specific activity $(k_1+k_2)/sa$ (g m^{-2} h^{-1}): A: Co, 420°C [38]; B: Mo, 385°C [81].

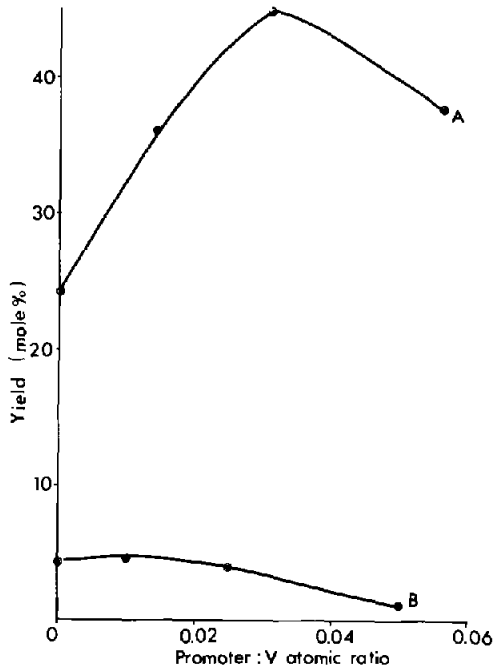


Fig. 4. Effect of promoters on maleic anhydride yield (mol-%): A: Fe [70]; B: Co [98].

in a way that reduced solid solution formation a decreased catalytic activity was observed. It is therefore concluded that the role of these promoters when effective at low concentrations is to form these solid solutions. In this respect it is interesting to note that a number of promoters form compounds with phosphorus that are isostructural to vanadium compounds and this probably aids solid solution formation, e.g. $TiHPO_4$, $ZrHPO_4$, [115,116], $MoOPO_4$ and $NbOPO_4$ [117].

A number of studies have shown that distinct optimum concentrations are observed for promoters. Representative data showing the effect on specific activity and yield (and hence selectivity) is shown in Figs. 3 and 4 respectively. It is clear that for different promoters a range of optimum concentrations is observed, as has been noted by Brutovsky and Gerej [67,68]. In addition, the same promoters can exhibit different optimal levels for different VPO compounds, e.g. for β-$VOPO_4$ $Co_{opt}/V=0.01$–0.02 [98], whereas for α-$VOPO_4$ $Co_{opt}/V=0.065$ [38]. The origin of these maxima is probably related to the formation of solid solutions and possibly represents the maximum concentration of the promoter that can form a solid solution with the VPO compound concerned. As noted by Hodnett and Delmon [98] additional levels of promoters usually lead to a distinct decrease in maleic anhydride yield. This effect would then be caused by the additional promoters forming separate phases, which under the oxidising conditions used would be expected to be oxides that

TABLE 5

Effect of n-butane concentration on maleic anhydride yield

No	Ref.	Company	Promoter	Promoter V[a]	Operating[β] Condition	Feed Composition (mol-%)					T (°C)	GHSV (h^{-1})	Conv. mol-%	Sel. mol-%	Yield mol-%	C_{max}	MA yield (mol MA l cat^{-1} h^{-1})	MA exit (mol %)
						nC_4H_{10}	O_2	N_2	CO	CO_2								
1	26	Chevron	none	-	Fuel Lean	1.5	20.7	76.8	0	0	374	1000	90	69	62	100	0.42	0.93
	81	ICI	Mo	0.04	Fuel Lean	1.5	20.7	76.8	0	0	340	300	85	82	70	100	0.14	1.05
	66	ICI	none	-	Fuel Lean	1.5	20.7	76.8	0	0	420	4000	62	68	42	100	1.13	0.63
2	120	Monsanto	none	-	Explosive Region	4.3	20.1	74.6	0	0	430	371	-	-	47	100	0.33	2.02
						3.0	20.4	75.6	0	0	423	366	-	-	57	100	0.28	1.71
3	121	Standard Oil	Zinc	0.19	Oxygen limited	3.4	9.3	87.3	0	0	430	1230	39	63	25	78.2	0.46	0.85
	132					3.4	11.0	85.6	0	0	430	1230	40	63	25	92.4	0.48	0.85
4	66,91	ICI	none	-	Fuel Rich	14.0	18.1	67.9	0	0	360	1100	17	78	13	36.9	0.91	1.86
			cerium	0.005	Fuel Rich	15.9	17.7	66.4	0	0	360	2260	14	71	10	31.8	1.59	1.58
			none	-	Fuel Rich	20.3	18.9	0	31.6	29.2	360	1000	9	80	7	26.6	0.65	1.46

[a]Atomic ratio, P/V = 1.0–1.2.

[β]See Fig. 5.

[γ]Maximum n-butane conversion at 100% maleic anhydride selectivity.

would promote only total oxidation. Hodnett and Delmon [98] have observed surface segregation of cobalt above the optimum level using the surface sensitive XPS technique.

The question that remains is by what mechanism do these solid solutions achieve the increased catalytic performance. However, there exists at present too few data which enable this question to be definitively answered. It has been noted that the phosphorus content of active catalysts is usually in excess of that required by the stoichiometry of the compound. This excess phosphorus enables maintenance of the optimum vanadium oxidation state, whilst preventing over-oxidation of the bulk catalyst structure. By formation of solid solutions this excess phosphorus would be incorporated into the structure and hence stabilised. As a consequence of this, one advantage of these solid solutions is possibly that they would be expected to be more resistant to phosphorus loss which is known to be a major problem for industrially operated catalysts. Hodnett and Delmon [98] in their study of cobalt promotion of β-$VOPO_4$ showed that the addition of cobalt significantly affected the reduction of the catalysts by stabilising the oxygen anions present in the structure. Hence, two further roles of the cations present in the solid solution could be (a) to stabilise an average vanadium oxidation state that would enable improved initial dehydrogenation of n-butane, and (b) to control oxygen diffusion and adsorption and thereby minimise the non-selective oxidation route. Evidence to confirm that promoters aid the control of the optimum oxidation state is found in the studies made above the upper explosion limit of n-butane/oxygen mixtures [66,91]. At these conditions, as noted by Hodnett [2], the promotional effect of lanthanum and molybdenum is less significant than when similar formulations are used with n-butane/air mixtures below the lower explosion limit. The reduced oxygen concentration, together with the increased n-butane concentration, probably regulates the average vanadium oxidation state, thereby decreasing the tendency for over-oxidation under these conditions. Hence, it is clear that further studies are required to fully elucidate the mechanism of action of promoters.

6. EFFECT OF n-BUTANE CONCENTRATION ON MALEIC ANHYDRIDE YIELD

The oxidation of n-butane to maleic anhydride is usually carried out using up to 1.5 mol-% n-butane in air, and by far the majority of studies, both in the academic and patent literature, utilise these gas compositions. These conditions are termed the fuel lean conditions since this is below the lower explosion limit which is shown schematically in Fig. 5. Very few studies have investigated reactant gas compositions other than fuel lean and representative data from these are summarised in Table 5. The fuel lean process (line 1, Table 5) is operated at high n-butane conversion (90%) and recovery of unreacted n-butane is not considered economic due to its low concentration in the exit gases.

This therefore represents a loss of about 10% of the n-butane fed to the reactor. Another serious disadvantage of fuel lean operation is that the exit concentration of maleic anhydride is at best about 1 mol-%. Since water vapour is co-produced in the partial oxidation process, at this low exit concentration, it is not usually considered feasible to condense significant quantities of maleic anhydride without co-condensing water, and hence, forming maleic acid. Since recovery of the acid has been inevitable, in practise the product is recovered by water scrubbing and the resulting maleic acid is subsequently dehydrated to the anhydride. Operation with higher n-butane feed concentrations has been found to overcome these major disadvantages.

Fig. 5 shows there are four distinct regions for n-butane/oxygen/inert gas compositions: (a) oxygen lean; (b) fuel rich; (c) explosion region, and (d) fuel lean. Initial studies with n-butane concentrations above the lower explosion limit used compositions within the explosion region, i.e. 3–5 mol-% n-butane in air using fluidised bed reactors [31,76,118,119]. Recently Kwentus and Suda [120] (line 2, Table 5) have utilised these compositions with a fixed bed reactor and have shown that hot spots, which can be a cause of low overall yields, can be successfully avoided if up 40 vol.-% inert granules are added to the first part of the catalyst bed. Operation in this regime produces high exit maleic anhydride concentrations (1.7–2.0 mol-%) a major part of which can be recovered as liquid maleic anhydride using conventional condensation methods. However, the yield of maleic anhydride in terms of mol l cat^{-1} h^{-1} is lower than that of the fuel lean processes, and hence, a larger catalyst volume would be required for operation with these catalysts under these conditions.

Frank [121] has studied operation of a V:P:Zn = 1:1.2:0.19 catalyst in the oxygen limited region (< 13 mol-%) using 3.4 mol-% n-butane as a reactant (line 3, Table 5). The equivalent catalyst has previously been well studied for fuel lean conditions [46] (line 2, Table 2). More recently Bither [78,79] has

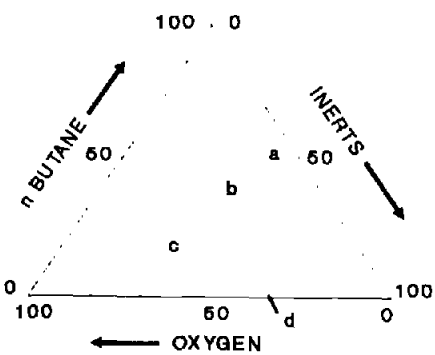


Fig. 5. Schematic diagram of regions for n-butane/oxygen/inert gas mixtures: (a) oxygen lean, (b) fuel rich, (c) explosion region, and (d) fuel lean.

also studied oxygen lean conditions. Frank disclosed a full scale design for the oxygen limited process which utilises a recycle to recover and react unconverted butane. In this way the low conversion per pass is not a significant disadvantage. Operating at lower conversions enables an increased selectivity to maleic anhydride to be achieved, and hence, the overall n-butane conversion efficiency is higher than that for the fuel lean process (oxygen limited process 0.94 g C_4H_{10} per g MA compared to 0.99 g C_4H_{10} per g MA for typical fuel lean process, stoichiometric 0.592 g C_4H_{10} per g MA). Some, about 50–60%, of the maleic anhydride in this process is recovered by condensation which is a further advantage and the remainder is recovered as maleic acid by conventional procedures. Unconverted n-butane, together with the total oxidation products carbon monoxide and carbon dioxide, are recovered and recycled. A purge of about 10% of the reactor effluent is made to ensure that inert gases do not build up to an unacceptable level. Although the n-butane efficiencies are improved with this process, the maleic anhydride yields are not significantly different from the fuel lean process and so no reduction in catalyst volume can be achieved with this regime. Centi et al. [122] have also studied n-butane oxidation under oxygen limited conditions in order to compare the mechanisms of n-butane and 1-butene oxidation. They concluded that the dehydrogenation of n-butane to 1-butene and 2-butene required a V^{IV}–V^{III} redox couple, whereas the subsequent oxidation of 1-butene and 2-butene to maleic anhydride, required a V^{V}–V^{IV} redox couple. They further considered that the mechanism of oxidation is similar at low and high n-butane concentrations, but there are differences in the surface distribution of the active sites, with the density of active sites being highest at low butane concentrations. When the feed consisted of 30.5 mol-% n-butane and 10.2 mol-% oxygen, 1-butene, 2-butene and 1,3-butadiene were observed at low levels when oxygen conversion was 100%. These partial oxidation products are not normally observed for n-butane oxidation, but are mechanistically significant, indicating that n-butane is sequentially converted to butenes and 1,3-butadiene prior to formation of maleic anhydride.

Operation above the upper explosion limit (region b, Fig. 5) has been exemplified by Higgins and Hutchings [66,91] using high surface area catalysts obtained by ball milling of the catalyst precursor (line 4, Table 5). These conditions are termed fuel rich and subsequently other workers have also studied butane oxidation under these conditions [123–126]. Dickason [123] demonstrated that increased selectivities to maleic anhydride could be achieved with the high n-butane feed concentrations, whereas Slinkard and Hughes [124] used a catalyst containing mainly V_2O_5 to co-produce maleic anhydride and acetic acid. Higgins and Hutchings [66,91] have shown that very high selectivities to maleic anhydride can be achieved under these conditions, i.e. up to 80 mol-% at 9% conversion. The reduced n-butane conversion due to the limitation in the oxygen concentration necessitates the use of a recycle process. Since carbon monoxide and carbon dioxide could build up in the recycle gases

and become a significant proportion of the feed gases, the effect of carbon monoxide and carbon dioxide on n-butane conversion was also studied and was not found to have a significant effect. Although the requirement for a recycle process is a disadvantage when compared to the currently operated single pass fuel lean process, operation in the fuel rich region has distinct advantages over the other three operating regions. First, it is possible to achieve increased maleic anhydride yields (80 mol-%) at decreased n-butane usages (typical fuel rich n-butane usage 0.76–0.81 g C_4H_{10} per g MA). Secondly, even at the low n-butane conversions per pass observed, the exit gases contain significantly higher maleic anhydride concentrations (1.5–2.0 mol-%) than other operating regions, and hence, a substantial yield can be recovered by condensation rather than using the conventional water scrubbing recovery technique. In this regime it is possible to operate at very high n-butane concentrations (e.g. 50 mol-%) and under such conditions the advantages outlined above could be expected to be enhanced. In addition, Blum et al. [127] have utilised a fuel rich activation procedure to improve the performance of a vanadium phosphorus catalyst in a fluidised bed butane oxidation process.

Operation in both the oxygen limited and fuel rich regions necessitates a recycle process. Bakshi et al. [128] have shown that in such recycle processes the concentrations of acids, that may be formed in the oxidation process, should be limited in the recycle stream to <100 ppm, higher levels are reported to significantly decrease both conversion and selectivity. Such a process step would necessitate the use of an aqueous caustic scrubbing following the recovery of maleic anhydride by condensation and water scrubbing. The mechanism by which the acids cause a loss of catalytic performance was, however, not elucidated and clearly requires further investigation.

7. CONCLUDING REMARKS

Analysis of the role of promoters in VPO catalysts for n-butane oxidation to maleic anhydride has identified that the major feature of most reported promotional effects is of a structural nature. A distinct role is apparent for these promoters in the attainment of the required precursor and final catalyst VPO phases. In addition, solid solutions of the type $((VO)_xM_{1-x})_2P_2O_7$, M = promoter, have been identified with very active catalysts, and the optimum concentrations observed for promoters are probably related to the maximum incorporation of the promoters into these solid solutions. However, it is clear that more systematic studies of promotional effects are now required to elucidate the possible role of promoters in the control of catalyst oxidation state and surface segregation of phases. Such studies could be carried out using well characterised solid solutions as model compounds for these effects. Although most of the work to date has involved promotion of $(VO)_2P_2O_7$,

α-$VOPO_4$ and β-$VOPO_4$, other compounds should now be studied in more detail, e.g. $VO(PO_3)_2$ suitably promoted may provide an active and selective catalyst.

The limited studies that have been made to date concerning the role of the butane/oxygen ratio have indicated that considerable advantage with respect to maleic anhydride yields can be obtained by increasing the n-butane concentration. The most promising of these approaches would appear to be in the fuel rich region and further studies of catalysts in this operating regime are required to consolidate the earlier work. The role of the promoter with increased n-butane concentrations requires evaluation, and again, well-defined solid solutions should be used as a basis for the evaluation.

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